

- 8 (a) This is to reduce collisions between protons and any gas molecules in the beam pipes, so that less harmful ionising radiation is produced.

OR

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that the proton beam can remain focused and not get scattered by these collisions.

OR

This is to reduce collisions between protons and any gas molecules in the beam pipes, so that there is less energy loss and the protons can reach the high speed required.

- (b) No. As neutrons are neutral and uncharged, their trajectories cannot be bent / curved by the presence of magnetic fields.

OR

No. As neutrons are neutral and uncharged, they cannot be accelerated by electric fields.

- (c) (i)

$$E = (\gamma - 1)m_0c^2$$

$$(7.0 \times 10^{12})(1.60 \times 10^{-19}) = (\gamma - 1)(1.67 \times 10^{-27})(3.00 \times 10^8)^2$$

$$1.12 \times 10^{-6} = (\gamma - 1)(1.67 \times 10^{-27})(3.00 \times 10^8)^2$$

$$\gamma = 7452.8$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = 7452.8$$

$$v = \sqrt{\left[1 - \left(\frac{1}{7452.8}\right)^2\right]} c^2 = 0.999999991 c \quad (\text{shown})$$

- (ii) Since beam current is defined as the average rate of flow of charge,

$$\text{beam current} = \frac{\text{total charge in the ring}}{\text{period}} = \frac{Q}{T}$$

$$T = \frac{\text{distance travelled along the ring}}{\text{speed of protons}} = \frac{26659}{0.999999991 c} = 8.8863 \times 10^{-5} \text{ s}$$

$$0.58 = \frac{Q}{8.8863 \times 10^{-5}}$$

$$Q = 5.1541 \times 10^{-5} \text{ C}$$

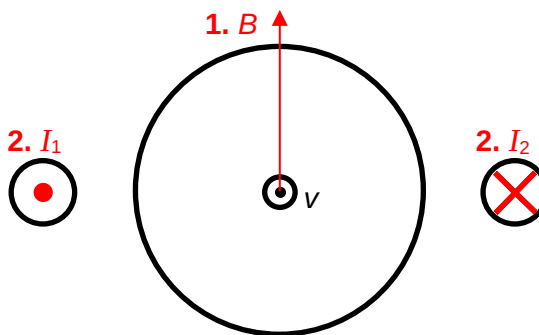
number of protons per proton beam,

$$N = \frac{Q}{e} = \frac{5.1541 \times 10^{-5}}{1.60 \times 10^{-19}} = 3.22 \times 10^{14}$$

Since there are 2808 bunches per proton beam,

$$\begin{aligned} \text{number of protons in each bunch} &= \frac{3.22 \times 10^{14}}{2808} \\ &= 1.1467 \times 10^{11} = 1.15 \times 10^{11} \quad (\text{shown}) \end{aligned}$$

(d) (i) 1.
 2.



1. Since proton velocity is out of the page, beam current is out of the page. Since centre of circular path of the proton's trajectory is on the left, the magnetic force on the proton that provides the centripetal force points to the left.
 Using Fleming's left-hand rule, direction of B is upwards.

*Straight line with arrowhead pointing upwards at the centre of pipe.

2. Since each combined cable carries current of the same magnitude, the direction of the current in each cable must be such that each produces a magnetic flux density that points upwards at the centre of the pipe, giving a resultant magnetic flux density as deduced in **(d)(i)1**.

Using the right-hand grip rule, I_1 points out of the page and I_2 points into the page.

*Correct directions, positions and symbols of both currents.

(ii) 1.

$$B = \frac{\mu_0 I_1}{2\pi d} + \frac{\mu_0 I_2}{2\pi d}$$

$$8.33 = \frac{(4\pi \times 10^{-7})(I_1 + I_2)}{2\pi \left(\frac{88}{2} \times 10^{-3}\right)}$$

$$I_1 + I_2 = 1832600 \text{ A}$$

Since $I_1 = I_2$,

$$I_1 = \frac{1832600}{2}$$

$$= 916300 \text{ A} = 9.16 \times 10^5 \text{ A}$$

2. Since current in each cable is 11850 A,
 $916300 = N(11850)$ where N is the number of cables
 $N = 77.3 \approx 77$

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3. Force on combined cable 2 by combined cable 1,

$$\begin{aligned}
 F &= B_1 I_2 L \\
 &= \frac{\mu_0 I_1}{2\pi d} I_2 L \\
 &= \frac{(4\pi \times 10^{-7})(916300)}{2\pi(88 \times 10^{-3})}(916300)(14.3) \\
 &= 2.7287 \times 10^7 = 2.73 \times 10^7 \text{ N}
 \end{aligned}$$

By Newton's third law, force on combined cable 1 by combined cable 2 is equal in magnitude and opposite in direction.

The cables in the dipole magnet setup experience an extremely large repulsive force away from each other. Hence, the stainless-steel collars are used to keep the cables from moving away from the beam pipe.

- (e) (i) average volume occupied by one proton

$$\begin{aligned}
 &= \frac{\text{volume of one bunch}}{\text{no. of protons in one bunch}} \\
 &= \frac{(7.48 \times 10^{-2})(1.0 \times 10^{-6})}{1.15 \times 10^{11}} \\
 &= 6.5043 \times 10^{-19} \text{ m}^3
 \end{aligned}$$

Approximating volume occupied by a proton to be volume of a cube with sides of length d :

$$d = \sqrt[3]{6.5043 \times 10^{-19}} = 8.6643 \times 10^{-7} \text{ m}$$

average electric force of repulsion between 2 protons,

$$\begin{aligned}
 F_E &= \frac{e^2}{4\pi\epsilon_0 d^2} \\
 &= \frac{(1.60 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(8.6643 \times 10^{-7})^2} \\
 &= 3.0663 \times 10^{-16} = 3.07 \times 10^{-16} \text{ N}
 \end{aligned}$$

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- (ii) Initial velocity v of proton is horizontal along the central axis of the beam pipe.

displacement of proton from centre of beam pipe to bottom of beam pipe,

$$s_y = u_y t + \frac{1}{2} g t^2$$

$$\frac{1}{2} (56 \times 10^{-3}) = 0 + \frac{1}{2} (9.81) t^2$$

$$t = 0.075554 \text{ s or } -0.075554 \text{ s (NA)}$$

total horizontal displacement travelled by proton,

$$s_x = vt$$

$$= (0.999999991)(3.00 \times 10^8)(0.075554)$$

$$= 22666199.8 \text{ m}$$

$$\begin{aligned} \text{number of rounds} &= \frac{s_x}{\text{circumference of ring}} \\ &= \frac{22666199.8}{26659} \\ &= 850.23 = 850 \end{aligned}$$

- (iii) The radiation produced from the accelerating protons will cause the protons to lose energy, resulting in the protons straying away from their original path as the radius of curvature of their trajectory will decrease.

Hence, the quadrupole magnets are needed to help keep the protons travelling along the central axis of the beam pipe.

(f) Advantages:

- Linear accelerators are less expensive to build as they do not require large numbers of electromagnets to keep the particles in a circular path, which need to be kept at extremely low temperatures with large amounts of liquid nitrogen and helium.
- Synchrotron radiation due to the particles accelerating is a lot less as particles are not constantly accelerating unlike when they are travelling in circular paths where they experience centripetal acceleration.
- Synchrotron radiation due to the particles accelerating is a lot less as particles are not constantly changing direction unlike when they are travelling in circular paths.

Disadvantages:

- The chances for collisions to happen is much lower because there is only one collision point in a linear accelerator, whereas there can be multiple collision points in a circular accelerator.
- Linear accelerators are not able to reach the same high energies as a circular accelerator without being unfeasibly long, because particles in a circular accelerator can circulate many times, getting boosts in energy many times before colliding.
- For linear accelerators to reach the same high energies as a circular accelerator they need to be extremely long which is very expensive to build.